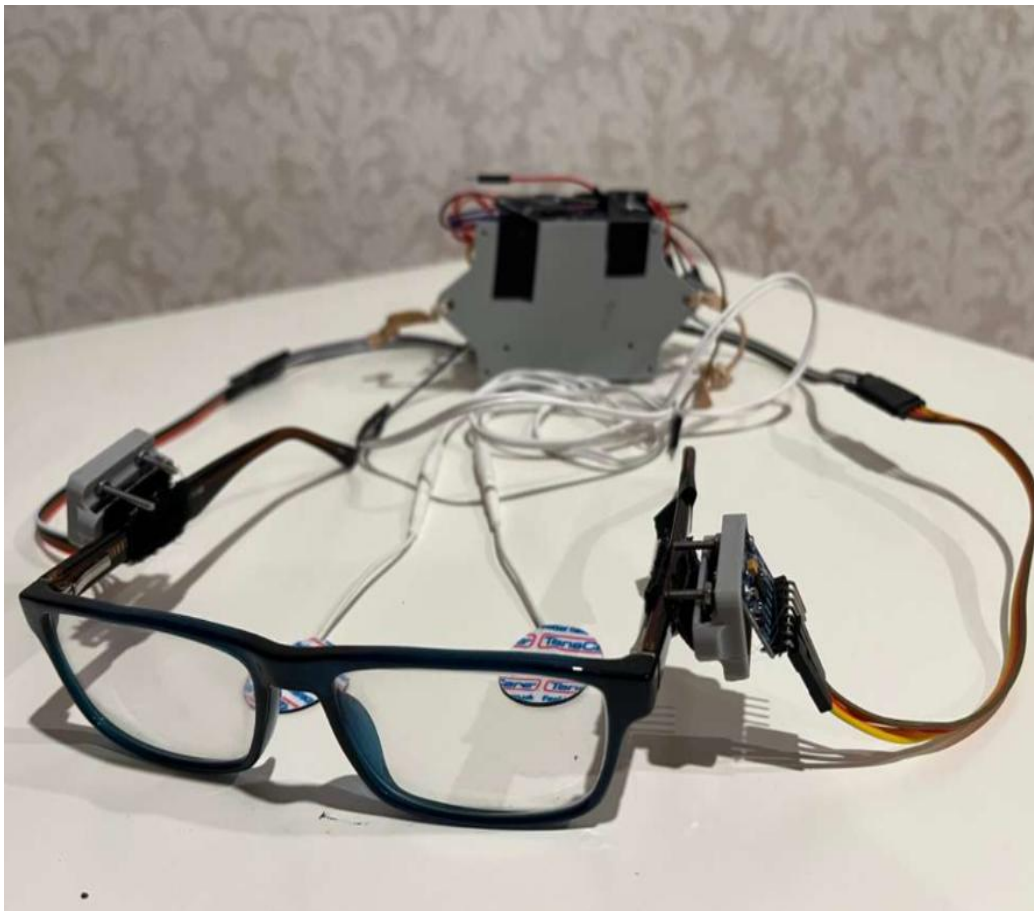


Statera: Adaptive Galvanic Vestibular Stimulation System for Fall Prevention



Final Design Study

Team ANTICx

Ishan Kasam, Tanay Mangal, Arnav Prabhudesai, Nichelle Thinagar, Chris Xu

Massachusetts Academy of Math and Science

May 21, 2026

Document Information

This document records how Team ANTICx designed, built, and tested Statera from the first ideas to the final prototype. Each section is written clearly enough that another student with a similar background could follow the logic and replicate the most important steps.

Section Writing and Editing Log

Section	Lead Writer	Editor	Written	Edited
Introduction and Background	Ishan Kasam	Tanay Mangal	5/7/26	5/14/26
Market Research and Requirements	Tanay Mangal	Arnav Prabhudesai	5/7/26	5/14/26
Preliminary Designs and Decision Matrix	Arnav Prabhudesai	Chris Xu	5/8/26	5/14/26
Build, Testing, and Prototyping Process	Chris Xu	Ishan Kasam	5/9/26	5/14/26
Design Studies	Nichelle Thinagar	Tanay Mangal	5/10/26	5/14/26
Final Device Summary and Future Work	Ishan Kasam	Nichelle Thinagar	5/12/26	5/14/26
Appendix and References	Tanay Mangal	All	5/13/26	5/14/26

Introduction

Motivation

Falls are one of the most serious health problems older adults face. A single fall can cause a broken bone, a head injury, a fear of moving, and a permanent loss of independence. The Centers for Disease Control and Prevention (2026) reports that more than 14 million older adults in the United States say they fall at least once each year, which works out to roughly one in four people over 65. What makes this especially dangerous is that falling once actually increases the chance of falling again, so waiting until after the first fall to act is already too late.

One cause of balance problems that often goes unaddressed is vestibular dysfunction. The vestibular system is the part of the inner ear that tells the brain how the body is tilted, how fast it is rotating, and where it is in space. Agrawal et al. (2009) found that vestibular dysfunction affects about 35.4% of adults over 40, which is roughly 69 million Americans. When the vestibular system sends the wrong signals, the brain gets bad information about balance, which makes falls more likely.

Galvanic vestibular stimulation (GVS) applies small electrical currents through surface electrodes placed near the mastoid bones behind the ears. Research has shown that GVS can influence the signals the vestibular system sends to the brain and can affect posture, balance, and spatial orientation (Marchand et al., 2025; Melo et al., 2022). Statera uses this approach by detecting early signs of instability with motion sensors and then delivering a controlled GVS signal that may help the user correct their balance before a full fall happens.

Problem Statement

There is no low-cost, wearable device available today that can detect early balance loss and deliver real-time vestibular feedback to help prevent falls before they happen.

Target Audience

Statera is designed primarily for people who have vestibular dysfunction or other balance-related conditions. This includes older adults in general, as well as people with specific diagnoses such as Meniere's disease, vestibular neuritis, bilateral vestibular hypofunction, Parkinson's disease, diabetic neuropathy, or acquired brain injury. The device is meant to support everyday activities like standing up, walking, turning, and transitioning from a seated to a standing position.

Purpose of this Document

This document will provide a detailed overview of the process that led to the final GVS-based fall prevention system, specifically covering the design studies used to evaluate the device, iterations made based on experiments, and a detailed analysis of the device and future steps to be taken.

Market Research Summary



Figure 1: Scandinavian Butler indoor rollator walker. Rollators provide mechanical balance support but are bulky, restricted to flat surfaces, and create dependency rather than rehabilitating the underlying vestibular deficit (Vandaele, 2017).



Figure 2: Standard single-tip walking cane. Canes redistribute weight but do not correct the underlying vestibular deficit, and are frequently misused without professional fitting (Wikipedia, 2024).



Figure 3: Apple Watch Series showing fall detection interface. The device detects falls after they occur rather than preventing them, and one study found a real-world detection rate of only 4.7% (McKnight's Senior Living, 2018).

Several device categories currently address balance and fall risk in older adults. Each carries meaningful limitations that motivate a GVS-based alternative. Walkers are the most widely deployed mobility aid for balance impairment. They are bulky, difficult to use on stairs, and replace rather than rehabilitate the user's vestibular function. Evidence exists for interference by walkers with natural balance and for impedance of lateral compensatory stepping movements, both of which can increase, rather than reduce, fall risk (Gell et al., 2015). Users who become reliant on walkers often find their mobility circumscribed rather than extended. (See Figure 1.)

Canes are much easier to carry and are seen much more commonly in the public compared to walkers, being able to easily fit into someone's life without drastically reducing the quality of it (Kayacan et al., 2017). Since canes don't need to be prescribed by a medical professional, users commonly decide themselves when they want to use a cane. However, that decision, when not informed by professionals, leads to actually increasing the risk of falls from not being a good fit or using it with the wrong technique (Physiopedia, 2021). Also, canes reduce the symmetry present while walking and can reduce both the cadence with how people walk and the length of their strides, making daily movements harder over time (Kayacan et al., 2017), which leads to the debate about canes and whether they help correct balance problems or they just make it worse in the long run (See Figure 2.)

The apple watch is almost at the other end of the spectrum as it is heavily reactive and digitalized as it detects falls and then it alerts emergency contacts. Through the use of several accelerometers, gyroscopes, and motion algorithms it is able to find abnormal and abrupt patterns in movement that match that of a fall alongside a verification period where if the user does not move for 60 seconds, it sounds an alarm and calls the contacts and emergency services (Retirement Living, 2025). However, Apple themselves state that the watch itself can't detect every fall and missing very gradual slips or falling from weird positions that it is not trained on (Funeral.com, 2026). WebMD detected that the Apple Watch was able to detect falls in the real world with only a 4.7% accuracy with a false-negative rate of 95.3% (WebMD, 2022). Apple watches are also expensive with Apple watches commonly ranging from \$199 to \$1,499 and additional monthly fees for circumstances like having a cellular emergency while calling if not paired with an iPhone (Betz et al., 2021). (See Figure 3.)

In contrast to all three device categories above, the proposed GVS device addresses the problem at its source. Rather than providing external mechanical support, replacing natural gait with device-dependent movement, or waiting to react after a fall, a GVS-based wearable modulates the vestibular signal itself, non-invasively, continuously, and without changing how the user moves through the world.

Preliminary Designs

Before arriving at the final design, the team considered four different design directions. This section describes each one with enough detail that another student could understand what components were involved, why each concept was considered, and why some were rejected.

Design Concept 1: GVS With IMU Sensing (Selected for Final Design)

This concept combined 4 inertial measurement units (IMUs) for motion sensing and fall detection, with the custom GVS circuit and electrodes for correcting posture. An IMU is a small chip that measures three-axis linear acceleration and three-axis angular rotation, and when the data from the IMUs show a movement pattern that looks like a fall, a microcontroller triggers the GVS output circuit and applies a current. The GVS circuit sends a small current through surface electrodes placed near the mastoid bones (the hard bumps directly behind each ear).

The main components of this design include 3 IMUs, 2 ESP32 microcontrollers, a custom circuit for the GVS system featuring a biphasic H-bridge built from NPN and PNP transistors for controlling electrode polarity, current-limiting resistors, TEMS surface electrodes, and a power supply. The GVS circuit and all its components are to be mounted on the back of a person's head on a glasses frame, and an additional necklace that holds one IMU will be included in order to provide more insight into a person's motion and prevent false positives. The initial concept sketch for this design can be observed in figure 1, featuring the ideas described above.

Design Concept 2: Balance Ballast System

This concept aimed to help prevent falls by shifting fluid from one side of the body to the other to counterbalance the user. The idea is similar to how ships

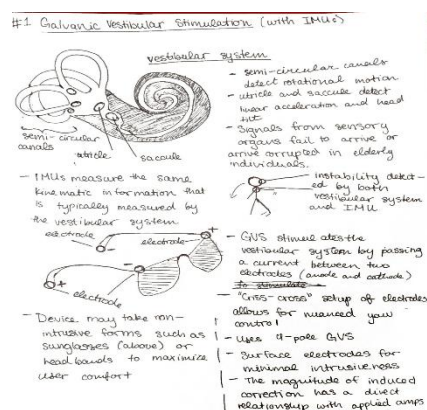


Figure 4. Earliest concept sketch for the GVS-with-IMU design. The sketch shows electrode placement near the mastoids and the IMU sensing approach.

and tightrope walkers stay balanced by redistributing weight. The system would use small fluid chambers attached to a body harness, pumps to move the fluid, IMUs to detect leaning, and a microcontroller to control the system.

Although the idea works in theory, it has several major problems for elderly users. The pumps would need to react very quickly, the fluid would make the system heavy, and the harness would likely be uncomfortable and difficult to put on. In addition, the fluid system would probably respond too slowly to stop a fall in time. Because of these issues, the concept was rejected early in development.

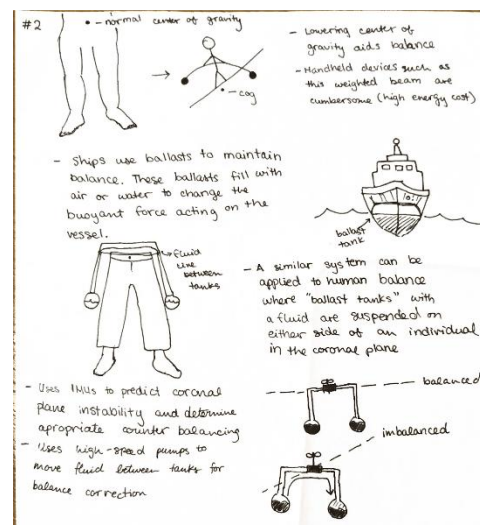


Figure 5. Balance ballast concept. Fluid would shift between side chambers to counterbalance a lean. Rejected due to weight, mechanical complexity, and slow response time.

Design Concept 3: GVS With Flex Sensors and Pressure Sensors

This concept ideally kept the GVS output from Concept 1 but replaced the IMUs with flex sensors mounted on the knee. Flex sensors are thin strips that change their electrical resistance in proportion to how much they are bent, and the idea was that the change in observed current could provide insight as to whether or not a fall was to happen. The plan was to place flex sensors on a knee sleeve to detect how much the knee was bending, and from there, inform the GVS system to decide the duration of current to be applied.

This concept was stopped early in development and was never integrated with the GVS system due to the high difficulty in detecting falls, due to the nature of the sensors themselves, along with the placement on the knee.



Figure 6: Flex sensor prototype placed on a knee sleeve during early testing. This design revealed that consistent sensor placement and wire management were major unsolved problems.

Design Concept 4: 3 IMU Based GVS Design

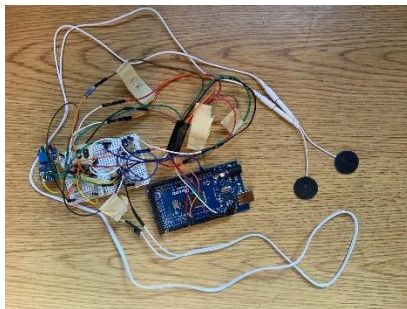


Figure 7. Full breadboard test platform used for initial circuit and firmware verification. This was not the final design but was critical for confirming the design worked before building the wearable form factor.

This concept uses three motion sensors mounted directly onto a pair of glasses, completely eliminating the need for a separate sensor necklace. By keeping all the components on the head, the device is much more comfortable, less intrusive, and significantly easier for people to put on and wear naturally every day.

While removing the necklace sensor slightly reduces the system's fall detection accuracy, the major improvement in overall daily comfort makes it a highly worthwhile trade-off. The entire setup—including the microcontroller, the three sensors, and the gentle electrical balance-correction circuit—is packed into one simple, self-contained wearable without any annoying extra accessories or wires.

Engineering Decision Matrix

The team used a weighted decision matrix to compare the three real competing design concepts. Design A is Concept 1 (GVS with IMUs). Design B is Concept 3 (GVS with flex sensors and FSRs). Design C is Concept 2 (the balance ballast system). Concept 4 (the breadboard test platform) was not a competing final design concept; it was a development tool used to test Concepts 1 and 3.

Wearability and Comfort received the highest weight (40 points) because a fall-prevention device is useless if the user refuses to wear it. Fall Prevention Effectiveness was weighted second-highest (30 points) because the device must actually help or the rest of the design does not matter.

Criteria (Max Points)	Level	Design A: GVS + IMUs	Design B: GVS + Flex/FSR	Design C: Ballast System	Notes
Wearability and Comfort (40)	L1	37	30	18	Ballast harness too heavy.
Fall Prevention Effectiveness (30)	L1	27	20	16	Flex only captures one joint.
User Safety (15)	L1	12	12	9	All require safety checks.
Noninvasive Design (15)	L2	15	15	12	Harness is borderline.
Mobility Preservation (15)	L2	15	11	9	Flex wires restrict legs.
Cost Accessibility (15)	L2	15	14	9	Pumps and chambers are expensive.
Usability and Appearance (15)	L3	15	11	7	Ballast looks unwearable.
Adjustable Assistance Level (10)	L3	10	8	5	
TOTAL (155 possible)		146	121	85	
TOPSIS Score		0.808	0.619	0.381	

Design A (GVS with IMUs) scored the highest overall with a TOPSIS score of 0.808. It was selected as the final design direction because it had the best combination of fall-prevention potential, wearability, safety, noninvasiveness, mobility preservation, and cost. The glasses-mounted form factor was developed during the later prototyping iterations to further improve comfort.

Prototyping Process and Design Iterations

This section explains the sequence of prototypes the team built and what each one taught the team. Every version addressed a specific problem found in the previous one. The descriptions are written as a cause-and-effect chain so that the final design can be understood as the result of real decisions made through testing, not just the first idea.

Iteration 1: Single IMU on Breadboard (Sensing Only)

The team ran movement tests: standing still, straight-line walking, turning 180 degrees, sitting down, and simulating a fall by leaning sharply to one side. The data clearly showed that fall-like movements produced large sudden spikes in combined acceleration magnitude that were visually different from normal walking. However, with only one IMU, the algorithm sometimes confused a sharp turn with a forward fall because both produced similar peak acceleration values.

This motivated the decision to add two more IMUs because three IMUs at different positions on the head give the algorithm more directions of data, which reduces the chance of a false positive from a fast normal movement.

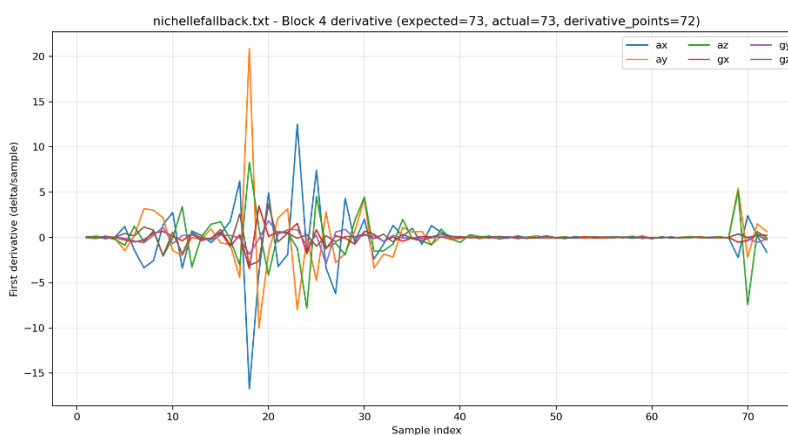


Figure 8: Example of raw data from one IMU during a backwards fall, clearly showing a change in signal when the fall occurs.

Iteration 2: Three-IMU Sensing System

Two more MPU-6050 modules were added, now there would be one IMU on either side of the head along with one on the back of the head. Because the Esp 8266 does not have enough ports to support all of the IMUs and the GVS system, 2 different microcontrollers were used, one dedicated for the IMUs, and one for the GVS system. The three IMUs were positioned at the center back, left temple, and right temple of the headband.

With three IMUs running, the classification algorithm compared combined readings across all three sensors simultaneously. A forward fall now produced a distinctive pattern across all three positions at once that was clearly different from any normal movement tested, and because the fall detection algorithm was based on thresholds, having multiple inputs where more thresholds would need to be satisfied increased accuracy. One issue with this design is that the mounting was slightly messy as there was no PCB, and there is a chance that people simply

moving their heads could be mistaken as a fall, as the IMUs do not have access to how the rest of the body is moving.

From this iteration, we made the decision to keep the three-IMU configuration but begin designing a compact glasses-mounted frame, as well as moving all of the electronics to a PCB for tighter integration. Also begin building the GVS output circuit separately so that both subsystems can eventually be combined. The ideal layout of this system is shown in figure 7.

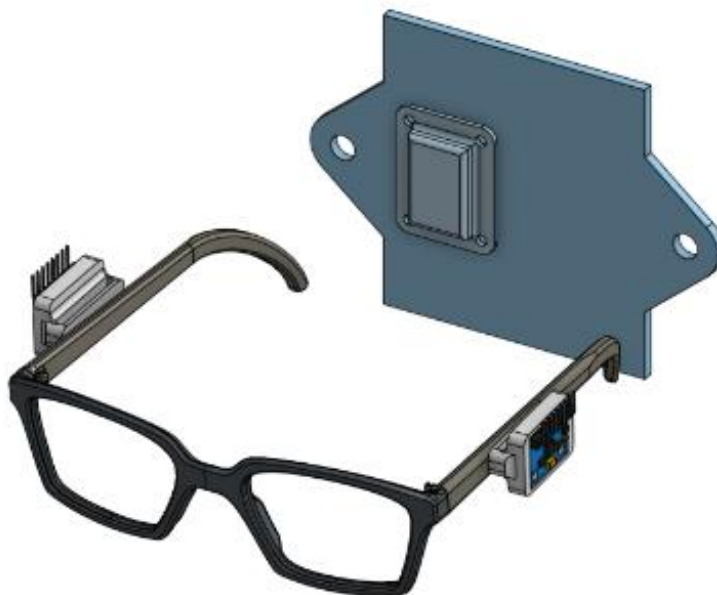


Figure 9: CAD model of the three-IMU glasses-based design with electronics mounting for the microcontrollers and all circuitries.

Iteration 4: GVS Circuit Without Boost Module (Not Felt)

With the three-IMU sensing confirmed to work somewhat working for fall detection, the team built the GVS output circuit on a separate section of the breadboard. The circuit, as previously described includes a biphasic H-bridge made from two 2N3904 NPN transistors and two 2N3906 PNP transistors. The two PNP transistors (Q1 and Q2) have their emitters connected to the positive supply rail. The two NPN transistors (Q3 and Q4) have their emitters connected to ground. Activating Q1 and Q4 together drives current through the electrodes in the forward direction. Activating Q2 and Q3 together drives current in the reverse direction. Heavy resistance is applied in order to make sure that a dangerous amount of current can never be applied.

In this first GVS build, the power supply was just the two AA batteries already powering the ESP32, which measured approximately 3.0 volts. The team checked the H-bridge operation with a multimeter and confirmed it was switching correctly. TENS lead wires were connected to 2-inch round cloth surface electrodes and placed near the mastoid bones behind the ears, and the multimeter read the ideal current for balance manipulation with GVS (0.3mA), but when tested on a human, there were consistent reports of not feeling any sensation. The reason for this is that there is an inherent resistance between human skin and the electrode, and so the voltage would

need to be higher in order for the current felt by the vestibular nerves to be sufficient for GVS to work properly.

This iteration motivated the decision to add a boost converter to raise the supply voltage because a higher voltage across the same skin resistance will produce more current. The MT3608 boost converter module was selected.

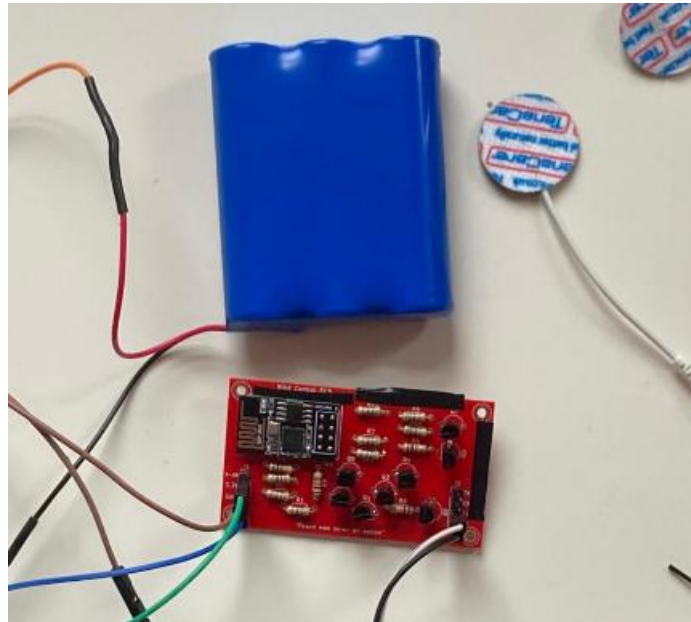


Figure 9: Iteration 4: GVS circuit running at 3.0 V from AA batteries only. The output current was too low to overcome skin resistance and no sensation was felt.

Iteration 5: GVS Circuit With MT3608 Boost Module (Output Felt)

The MT3608 boost converter was added between the LiPo battery pack and the H-bridge circuit to increase the voltage. The battery connected to the input side of the boost converter, and the output side connected to the H-bridge power lines. The sensing system, which included the ESP32, IMUs, and multiplexer, continued using a separate 3.3-volt power line from the ESP32 regulator. Both systems shared the same ground connection.

The team used a multimeter to measure the boost converter output voltage. The small adjustment knob on the module was slowly turned until the output reached 12.0 volts. The setting was then secured with a small drop of super glue so it would not move accidentally. At 12.0 volts and an estimated skin resistance of 50,000 ohms, the expected current was about 180 microamps, which is high enough for a person to feel the stimulation. To improve safety, an additional 1 kilohm resistor was added in series with each electrode wire to limit the maximum possible current if skin resistance suddenly decreased.

When the system was tested at 12.0 volts with the electrodes placed correctly behind the ears, the tester immediately felt a mild tingling sensation near the mastoid area. This matched the expected sensation for low-level GVS stimulation. No discomfort, dizziness, or nausea was reported during testing. The test was stopped immediately anytime the tester reported discomfort. This was the first successful test of the GVS subsystem.

After this success, the team decided to keep the MT3608-powered H-bridge design as the final GVS output circuit. The next step was to combine the sensing and stimulation systems and begin designing the final PCB layout for the glasses-mounted prototype.

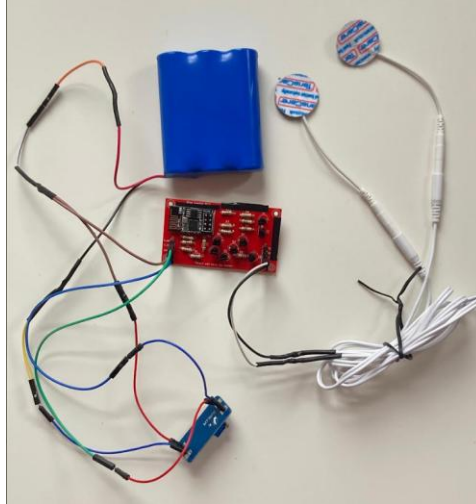


Figure 10: MT3608 boost module stepping AA battery voltage up to 12.0 V. This was the first iteration in which the GVS output was confirmed felt by the tester.

Iteration 6: Final Circuit Arrangement with IMUs and GVS System

The three-IMU sensing circuit and the boost-powered GVS circuit were combined onto one breadboard. The firmware was updated so the microcontroller reads all three IMUs through the multiplexer, runs the fall classification algorithm, and activates the H-bridge only when a fall-like event is detected and the safety timer has not been exceeded. The full closed-loop system was tested end-to-end for the first time: IMU detects fall simulation, algorithm classifies it, GVS output is felt within approximately one second.

After confirming the integrated breadboard system worked, the team designed a PCB layout to replace the loose wiring. The PCB brings the transistors, resistors, and component footprints together into a compact board that fits on the glasses front plate.

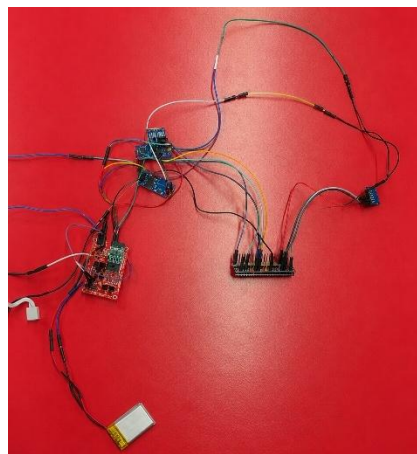


Figure 11. PCB layout for the integrated sensing and stimulation circuit. This design is intended to mount directly on the glasses back plate, and the IMUs can mount to the sides of the glasses.

Iteration 7: Glasses-Mounted Form Factor (Final Prototype)

The final form factor is a 3D-printed glasses frame with a front electronics mounting plate, two side brackets that hold one IMU module each, and a channel along each glasses arm for routing the electrode wires toward the ears. The battery and boost module are housed in a small enclosure that attaches to the frame. The third IMU mounts on the front plate at center forehead.

This iteration eliminates the headband and its associated loose wires. The glasses frame keeps all sensors in a fixed position relative to the head, which improves sensor placement consistency compared to Iteration 3 and makes the comfort scores significantly higher than any previous version.

This is the prototype that was submitted at the Testing Readiness Review and used for all three design studies.

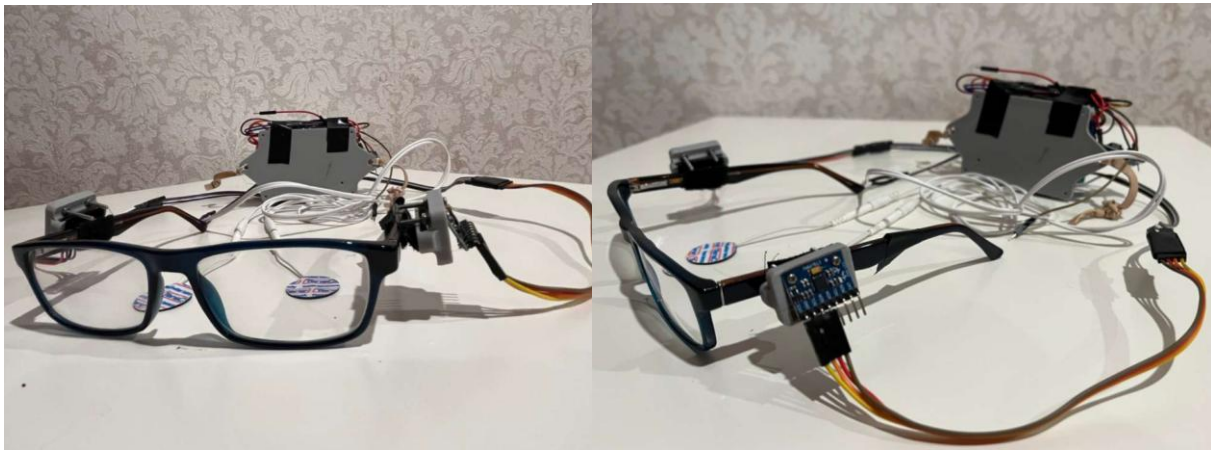


Figure 12. Glasses-mounted form factor. Front view (left) and side view (right) showing electrode wire routing and IMU placement.

Build and Testing

The build steps below describe how to duplicate the final Iteration 7 prototype. They are written as main steps with enough detail that a student with basic experience in Arduino, breadboards, and 3D printing could follow along. Earlier iterations that were abandoned are documented in the Prototyping Process section above.

Step 1: Design and Manufacture Electronics Mounting

Export each of the following parts from the CAD file as individual STL files: the glasses back PCB housing, and back housing for battery and IMU, the left side IMU mount, the right side IMU mount, and the battery and boost module enclosure. Print all parts in PETG filament at 0.2 mm layer height, 20% infill, with supports enabled for all overhanging features.

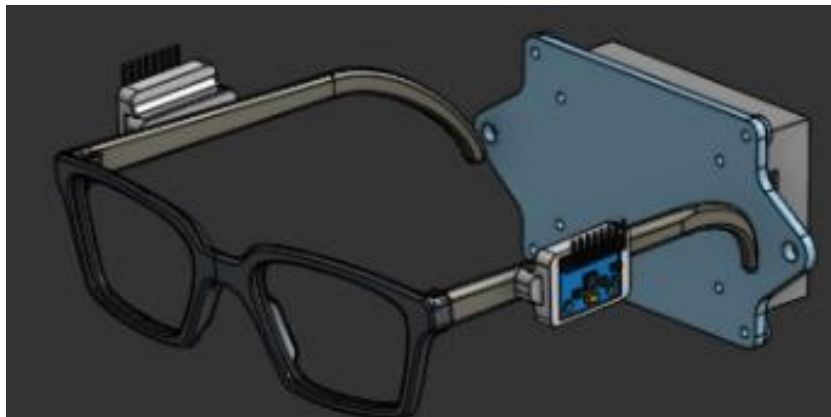


Figure 13: CAD layout from OnShape featuring 3D printable mounts for the IMUs as well as a back plate for mounting the GVS PCB, power source and MCU.

Step 2: Wire the Three IMUs

Connect the three MPU-6050 IMU modules to the ESP32 by sharing certain grounds and power sources to ensure that each IMU still functions. The AD0 pin on each IMU can be left floating or pulled to GND (address 0x68) since the multiplexer isolates them. After wiring, mount IMU 1 on the front plate center and IMUs 2 and 3 on the left and right-side mounts using M2 screws or a small drop of epoxy. Route the four-wire cable (VCC, GND, SDA, SCL) for each IMU through the printed channels in the frame to the front plate where the ESP32 and multiplexer sit.

Step 3: Build the GVS H-Bridge Circuit

The GVS circuit uses two 2N3906 PNP transistors (Q1 and Q2) and two 2N3904 NPN transistors (Q3 and Q4) in an H-bridge configuration. Connect the emitters of Q1 and Q2 to the MT3608 boost module output (12 V rail). Connect the emitters of Q3 and Q4 to ground. The collector of Q1 and the collector of Q3 both connect to the left electrode output terminal. The collector of Q2 and the collector of Q4 both connect to the right electrode output terminal.

Forward stimulation (left-to-right current): ESP32 drives Q1 base through a 10 kilohm resistor and drives Q4 base through a 10 kilohm resistor. Reverse stimulation: ESP32 drives Q2 and Q3

bases. Never drive Q1 and Q2 high at the same time or Q3 and Q4 high at the same time, as this creates a short circuit across the 12 V rail. Place a 1 kilohm safety resistor in series with each electrode output terminal to limit the maximum current that can flow through the skin.

Before any human contact, test the circuit by bridging the two electrode terminals with a 50 kilohm resistor to simulate skin. Measure the current through the resistor with a multimeter in series. It should read between 0.05 and 0.20 milliamps at 12 V output.

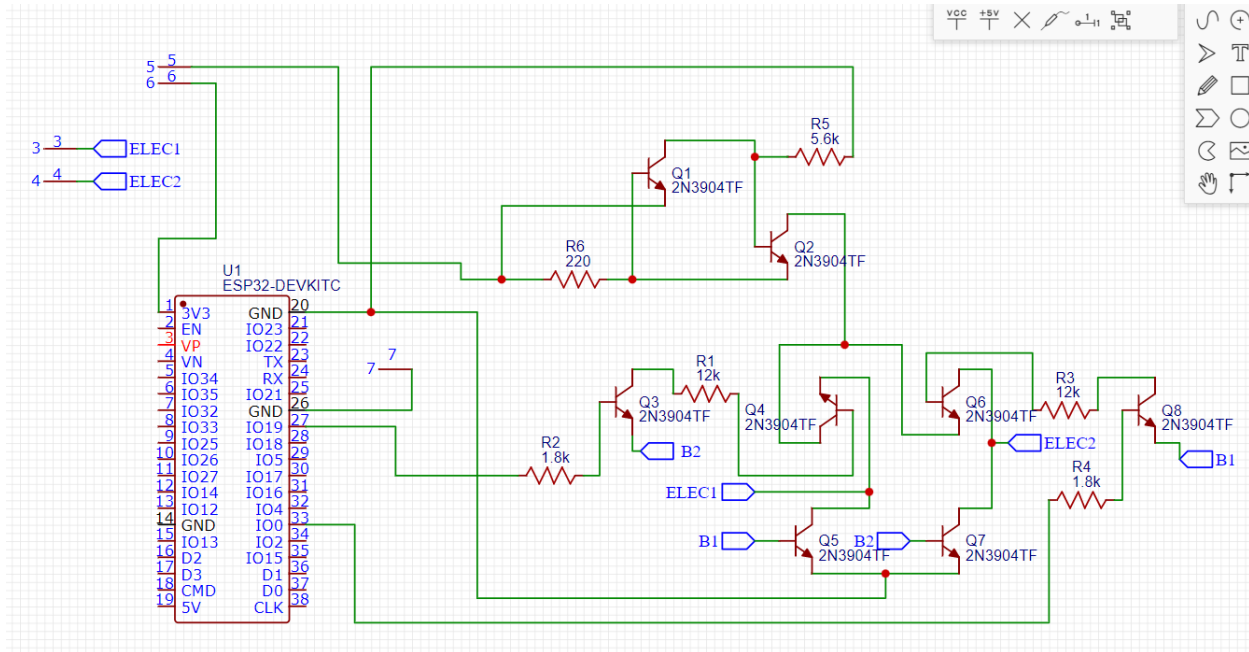


Figure 14: Full schematic of the custom PCB supporting an ESP 32 DevKit for the MCU, ports for 3 IMUs, as well as the full GVS setup with the biphasic H-bridge and adequate resistance to reach the desired current.

Step 4: Set the MT3608 Boost Module to 12 Volts

Connect the MT3608 input (IN+ and IN-) to the AA battery pack. Do not connect a load yet. Power on the battery pack and use a multimeter to measure the MT3608 output voltage. Turn the small onboard potentiometer clockwise in small increments. Stop when the multimeter reads 12 V. Mark the potentiometer position with a small dot of nail polish so the setting cannot drift accidentally.

Important: keep the GVS 12 V power rail completely separate from the ESP32 3.3 V sensing rail. Connect them at a single common ground point only. Mixing the two rails risks damaging the ESP32 or the IMUs.

Step 5: Upload Firmware and Verify Sensor Output

Open the Arduino IDE. Install the Adafruit MPU6050 library from the library manager. Select the correct ESP32 board from the boards manager. Upload the IMU data collection sketch first. Open the serial monitor at 115200 baud. While holding the prototype still with the front plate level, the Z-axis acceleration should read approximately 9.8 m/s-squared and the X and Y

axes should read near zero. If all three IMUs are returning reasonable values, the multiplexer wiring is correct. Finally, upload the full integrated firmware.

Step 6: Apply Electrodes and Run Safety Checks Before Every Session

Use 2-inch round white cloth surface electrodes. Clean the skin over the mastoid bone (directly behind each ear) with an alcohol wipe and let it dry for at least 60 seconds. Attach the electrode to the cleaned skin, pressing firmly around the edges to ensure full contact. Clip the TENS lead wire connectors to the electrode snaps before placing the electrodes on skin.

Safety checklist to run before every test session: (1) confirm MT3608 output is still 12 V with a multimeter, (2) verify the H-bridge by measuring current through the 50 kilohm dummy resistor, (3) confirm the serial monitor shows valid IMU values from all three sensors, (4) confirm the 500-millisecond safety shutoff is active in the firmware, (5) have a second person present who can power off the device immediately if needed. Do not skip any of these steps.



Figure 15. Completed Iteration 7 prototype worn during a test session. Electrodes are placed on the mastoid bones behind each ear. The glasses frame holds all components in position without loose wires.

Design Studies

Three design studies were conducted. The first design study determines whether the IMU system can actually classify falls. Study II determines whether the GVS circuit produces a real, detectable output. Study III determines whether the design became measurably more comfortable across the iteration process.

Design Study I: IMU Fall Detection Accuracy

Purpose

This test was used to determine whether the three-IMU system and final version of the fall classification algorithm can correctly identify fall-like movements as falls, and correctly identify non-fall movements as non-falls, with a statistically meaningful level of accuracy.

Independent Variable

The independent variable was the type of movement performed by the tester. Fall-like trials included: forward fall simulation, backward fall simulation, and sharp lateral lean beyond 45 degrees. Non-fall trials included: normal forward walking, sit-to-stand transition, turning 180 degrees in place, and moderate side-to-side weight shifting.

Dependent Variable

The dependent variable was the output of the algorithm for each trial, recording whether an impending fall was detected, and the severity of the balance loss event. The actual movement type was independently labeled by a team member watching the trial. Accuracy is computed by comparing the algorithm's output to the true label.

Controlled Variables

- Same tester for all 50 trials, to eliminate variation in body size and movement style.
- Same sensor placement, code, and overall device setup.
- Same firmware version with no code changes between trials.

Methodology

1. Record the true label of the upcoming trial type, fall or non-fall, as well as the specific movement, before performing the test.
2. Tester falls onto the gym mat in the specified direction.
3. Record the algorithm's output immediately after the trial ends.
4. Run 22 fall-like trials and 28 non-fall trials (50 total), with the type randomized to prevent order effects.

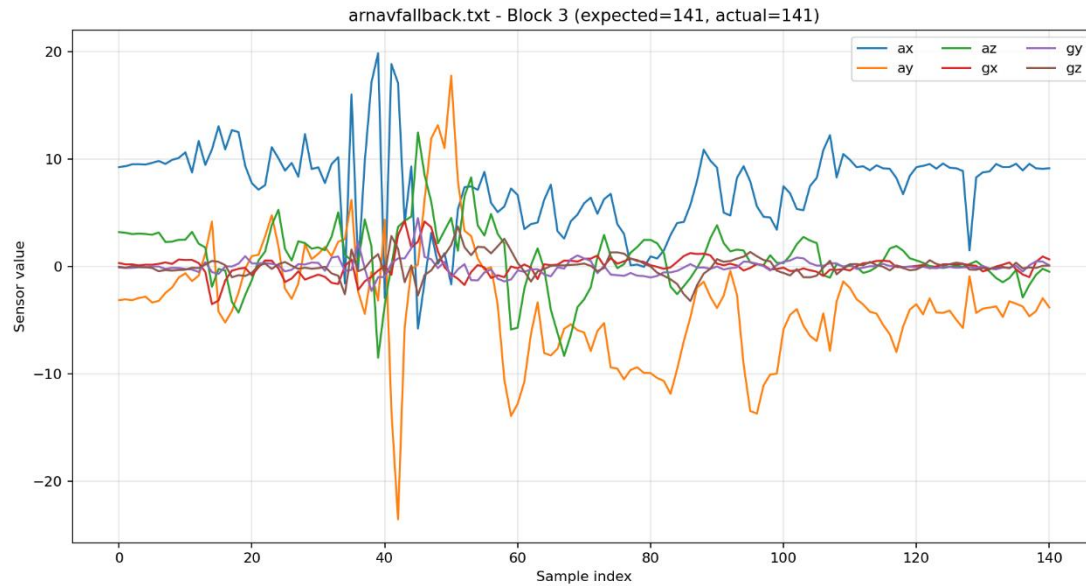


Figure 16. Example IMU acceleration magnitude vs. time for one backward-fall trial. The large spike corresponds to the fall motion and the near-zero region corresponds to the impact. This signal pattern is what the algorithm uses to classify falls.

Results

The confusion matrix below shows the results of all 50 trials.

	Predicted: Fall	Predicted: Non-Fall	Row Total
Actual: Fall =	18	4	22
Actual: Non-Fall	2	26	28
Column Total	20	30	50

Table DS1. Confusion matrix for the IMU fall detection study. 50 total trials: 22 fall-like and 28 non-fall.

The algorithm correctly classified 44 out of 50 trials, giving an observed accuracy of 88%. There were 4 false negatives (fall movements classified as non-fall) and 2 false positives (non-fall movements classified as fall).

Statistical Analysis

Analysis and Conclusion

This experimentation shows that the algorithm is mostly competent in differentiating balance loss events from basic movements, as can be seen with the 88% accuracy and statistically significant results with a p-value of $p < 0.001$. One limitation of the current algorithm is that it especially struggles with classifying the event of the transition between sitting

and standing, as this movement was seen to trigger more false positives than any other one. The current fall detection algorithm was also highly proficient in its low latency, which was not included in the purpose of this design study, but the average time between the onset of the balance loss event, and the algorithm's output time being 200ms is a positive result.

88% accuracy is a promising result for a minimum viable product at this stage, and the algorithm is not yet finished, but it works significantly better than chance and performs well on the most common fall types. The next step is testing with more testers and more movement variety.

Design Study II: GVS Circuit Output Detectability

Purpose

To determine whether the GVS circuit produces an output that the user can actually feel, and to verify that the reported sensation is caused by electrical stimulation and not by the physical pressure of the electrode on the skin or some sort of placebo effect.

Independent Variable

The independent variable was the duration of the application of the current from the GVS system, ranging from 0 seconds to 0.5 seconds.

Dependent Variable

Whether the tester reported feeling any sensation within a 5-second response window after each trial start (yes or no). For every 'yes' response, the tester also reported the approximate location and rated the intensity on a scale from 1 (barely noticeable) to 10 (uncomfortably strong).

Controlled Variables

- Electrode placement and GVS system remained unchanged.
- Skin cleaned with alcohol wipe and dried before each electrode application.
- The test subject was kept unaware of the oncoming current application throughout the entirety of the design study.

Methodology

1. The current output of the GVS system was verified prior to applying the electrodes.
2. The tester's skin was cleaned with an alcohol wipe and then set to wait for 60 seconds.
3. The electrodes were attached to the mastoid processes and grounded at the temples.
4. Explain to the tester: some trials will be real stimulation, some will not. Report honestly what is felt after each trial.
5. The operator randomly selects 25 active and 10 fake trials and records the order on a sheet the tester cannot see.
6. For each trial: operator either activates the circuit (active) or does nothing (sham). After 5 seconds, ask: 'Did you feel anything? If yes, where and how strong (1-10)?'. Additional comments will also be collected.
7. Record the tester's response and the operator's chosen condition for each trial.

Results

	Shock Felt	Shock Not felt	Row Total
Shock Delivered	25	0	25
Shock Not delivered	3	7	10
Column Total	28	7	35

Table DS2. Confusion matrix for the GVS detectability results. Active trials used 12 V stimulation; sham trials had the circuit off. The sham condition verifies that reported sensations are caused by the electrical output and not by electrode pressure alone.

Statistical Analysis

Analysis and Conclusion

The results showed that participants detected sensations during 100% of active trials compared to only 30% of sham trials, confirming that the GVS circuit produced a real and detectable electrical effect rather than sensations caused by electrode pressure or placebo effects. The lower sham detection rate also demonstrated that the single-blind methodology was effective at reducing participant bias.

One major takeaway from this study was that the GVS system is capable of consistently delivering current that caused noticeable balance changes. Based on these results, the next stage of development will focus on mounting the device such that electrode placement and electronics remain consistent and stable so that the GVS system can continue to function properly.

Design Study III: Comfort Across Prototype Iterations

Purpose

To determine whether the design iterations measurably improved comfort, invasiveness, and stability compared to earlier versions, and to confirm that the final glasses-mounted design is the most wearable option the team produced.

Independent Variable

The prototype version being worn: Design 1 (flex sensor knee sleeve), Design 2 (single-IMU headband), Design 3 (three-IMU glasses design), and Design 4 (PCB glasses-mounted final design).

Dependent Variable

Three scores rated on a 1-to-10 integer scale by three independent raters: Skin Comfort (1 correlates to uncomfortable contact with skin, and 10 correlates to no discomfort), Invasiveness (1 correlates to severely restricts natural movement, and 10 correlates to not restricting movement at all), and Stability (1 = device does not stay in place, 10 = stays in place securely during all activity). The reported value for each category is the mean of the three raters' scores.

Controlled Variables

- Three independent raters for every prototype version. Raters were team members who had not been the lead builder for that prototype version, to reduce familiarity bias.
- Same activity sequence for every prototype version: stand for 30 seconds, walk 10 steps forward and 10 steps back, sit down, stand up.
- Same rating scale definition sheet given to all raters in writing before scoring began.
- Raters submitted their scores privately before seeing any other rater's scores.

Methodology

1. Distribute the rating scale definition sheet to all three raters and go through definitions together to confirm consistent interpretation.
2. Subject puts on the prototype version and completes the standard activity sequence.
3. Each rater privately records Skin Comfort, Invasiveness, and Stability scores on a paper form.
4. Collect all three forms and calculate the mean score for each category.
5. Repeat for all four prototype versions. Order is randomized across sessions.

Results

Prototype Version	Skin Comfort (1-10)	Invasiveness (1-10)	Stability (1-10)	Mean Score
Design 1: Flex Sensor Knee Sleeve	4.2	4.8	4.6	4.53

Design 2: Single-IMU Headband	8	8.4	8	8.13
Design 3: Three-IMU Glasses	7.8	8	7.6	7.8
Design 4: PCB Glasses (Final)	8	8.2	7.8	8

Table DS3. Average comfort scores across three raters for each prototype version. Scale: 1-10, where 10 is best. Each value is the mean of three independent rater scores.

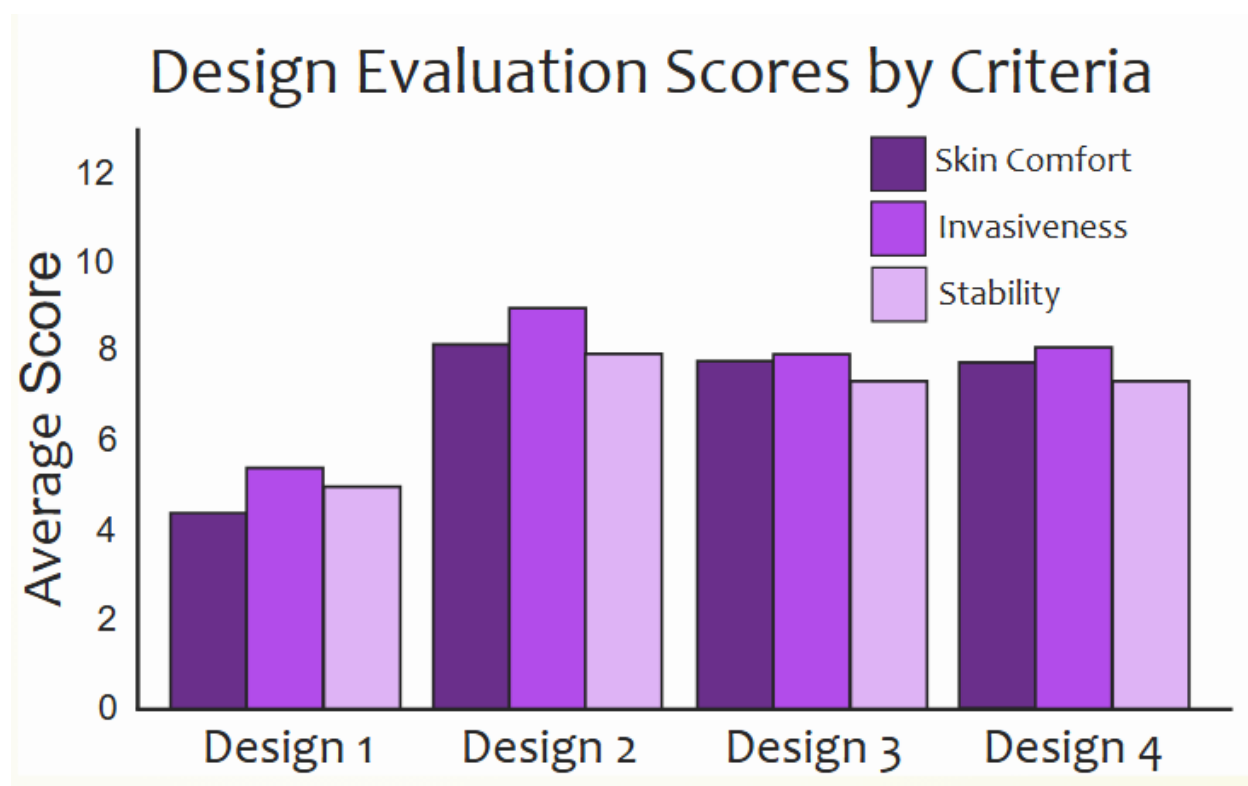


Figure 18. Grouped bar chart of average comfort scores across all four prototype versions. Iteration 7 (the final glasses-mounted design) scored highest in all three categories. Source: Team ANTICx design study session, 2026.

Statistical Analysis

Analysis and Conclusion

The final PCB glasses design scored near the top overall and was the strongest fully integrated prototype because it combined comfort, stability, sensing, and GVS functionality in one wearable form factor. The main weakness still visible in the glasses design is electrode adhesion: if the electrode pads lift during vigorous activity, the Stability score would likely drop. This finding directly motivated the electrode improvement steps listed in the Future Work section.

What Works and What Does Not Work

What Works

- The final glasses-mounted device, featuring a PCB enclosure and consolidated electronics integrated directly onto the frame, provides stable and secure mounting that keeps the components protected and safe for the user while maintaining a relatively sleek form factor. The system is also easy to use and scores highly for comfort, requiring only that the user wear the glasses and apply two discrete electrodes.
- The three-IMU sensing system, paired with the current fall-detection algorithm, provides meaningful insight into whether GVS intervention is necessary and successfully distinguished fall-like movements from non-fall movements in 88% of 50 controlled trials. The 88% classification accuracy is statistically significantly better than random chance, confirmed by chi-square test $\chi^2(1) = 28.63$, $p < 0.001$, with a 95% confidence interval of [0.762, 0.944].
- The GVS circuit with an H-Bridge and voltage boost module produces a detectable stimulation sensation in 100% of active trials, confirmed as statistically different from the 30% sham detection rate $\chi^2(1) = 21.88$, $p < 0.001$.
- The design is truly noninvasive, as it uses only external surface electrodes and external sensors, and the device will not intrude upon people's everyday lives in the same way a cane or walker might.
- • The final prototype cost approximately \$69.02 in purchased components, remaining well below the \$150 Level 1 requirement and demonstrating strong potential for affordability and practical feasibility for the target user population.

What Does Not Fully Work Yet

- The fall detection algorithm produces 4 false negatives from slow lateral leans and 2 false positives from fast sit-to-stand transitions. More labeled training data is needed for these edge cases and for classifying a wider range of movements.
- The electrodes are not completely stable in their mounting, and they require the skin to be cleaned and dried prior to application in order to stay adhered to the mastoid processes for more than the time of one testing trial.
- • The current prototype could be further refined through improved component mounting and wire management, as the relatively large enclosure positioned at the back of the head can cause minor discomfort during use.

Final Device Summary

The final prototype is a glasses-mounted, closed-loop wearable system that can detect and prevent falls due to minor balance loss events in real time. Three IMUs located around the head continuously monitor head motion and send data to a microcontroller. When the fall classification algorithm detects a fall-like movement, it activates the GVS output circuit, which sends a controlled electrical current through surface electrodes near the mastoid bones to manipulate the body's balance system into correcting itself and ultimately prevent the fall.

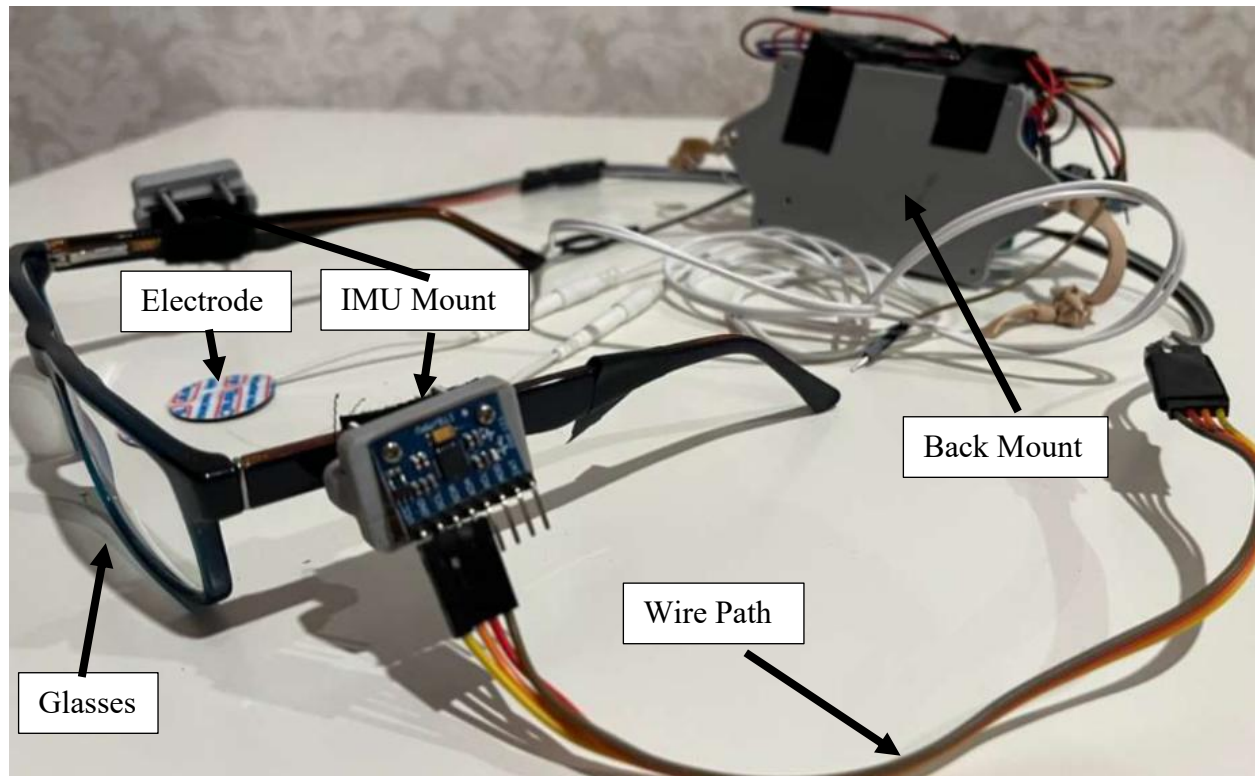


Figure 20. Final Statera prototype, side view. Wire routing along the glasses arm keeps the cable management clean.

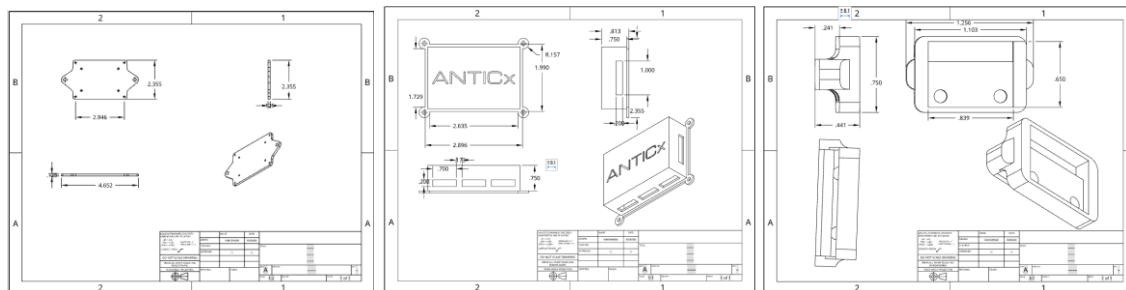


Figure 21. Dimensioned CAD drawing of the final glasses-mounted prototype. All dimensions are in inches.

Level 1 Requirement Status

The table below gives an honest assessment of each Level 1 requirement. Requirements that are partially met are explained so the reader understands exactly what is still needed.

Level 1 Requirement	Met?	Explanation
Reduce balance loss events.	Partial	IMU classifier detected fall-like motion at 88% accuracy in controlled trials. Real-world fall reduction in target users has not been tested.
No physiological harm.	Yes	Electrical safety verified by multimeter at every session, and literature backs the idea that GVS is safe for consistent usage. Clinical safety validation is future work.
Noninvasive.	Yes	External surface electrodes and external sensors only.
Does not impede mobility.	Yes	Glasses based mounting is hands-free, does not restrict gait, does not limit head movement and is additionally not very bulky.
Fast enough response.	Partial	Classification was demonstrated, and balance was manipulated before a fall happened but exact latency from movement onset to GVS output has not been directly measured yet.
Cost under \$150.	Yes	Purchased component cost: approximately \$65.20.

Future Work

The primary future work for this project revolves around refining already existing elements, broadening the scope of the functionality of the device, and also expanding the testing and feedback as well as collaborating with members of the target audience to further this device.

One of the most important areas for future work is the fall detection algorithm, as it primarily needs a larger and more diverse dataset to support a more sophisticated movement classification system and ultimately become reliable enough for real-world use. While the current system was able to classify a good amount of movements as falls and correctly refrain from identifying some non-falls as non-falls, we found that IMU readings of falls can vary widely in nature just because there are many different types of movements that people can make that can be mistaken for falls, an example being the transition from sitting to standing.

The electrode system is another problem to solve, as the current adhesive cloth electrodes lift during extended wear, which ends up breaking the GVS circuit entirely. Future iterations can focus on improving the adhesive used by the electrodes or aim to implement a new mechanical system that holds the electrodes in place.

Additional improvements to the design in terms of comfort, aesthetics, and reduced invasiveness include consolidating all electronics onto a single central printed circuit board, rather than relying on the jumper wires and pin connections currently used for prototyping and iterative development.

Finally, everything done so far has been tested on healthy young adults simulating falls. The only meaningful next step for validating whether Statera actually works is a study involving people who have real vestibular dysfunction, which is the population the device is designed for.

Appendix A: Bill of Materials and Tools

The components listed below represent the purchased parts for the final Iteration 7 prototype. Note that the MPU-6050 quantity has been corrected to three units to match the actual final design (the original grant proposal listed one). All prices are as purchased; current prices may differ. Items marked as on hand were already available in the lab.

Component	Quantity	Unit Price	Total	Order Links
2N3906	1	7.99	7.99	Amazon.com: 2n3906
2N3904	1	7.59	7.59	Amazon.com: 2n3904
MPU-6050	3	11.79	11.79	MPU 6050
ESP-01v090	2	6.59	13.18	Amazon.com: DIYmall ESP8266 ESP-01S WiFi Serial Transceiver Module with 4MB Flash
MT3608 Boost Module	1	6.99	6.99	Voltage Regulator
2" Round White Cloth Electrodes (TYCO Gel)	3	2.5	7.5	Amazon.com: TensCare 1" Circular TENS Unit Replacement Pads,
Roscoe Medical TENS Lead Wires (2mm Pin)	2	6.99	13.98	Lead Wires and TENS EMS Units
Total			69.02	

Tools Used

Tool	Purpose
Arduino IDE	Writing, compiling, and uploading firmware to the ESP32. Also used for the serial monitor and serial plotter during testing.
CAD software (OnShape)	Designing the glasses frame, electronics plate, side IMU mounts, and battery enclosure for 3D printing.
BambuLab H2D	Printing all prototype housing components.
Digital multimeter	Verifying output voltage, checking H-bridge behavior with dummy resistor, confirming circuit continuity, measuring current through electrodes.
Soldering iron and solder	Making permanent connections on the PCB and terminating electrode lead wires.
Breadboard and jumper wires	Rapid circuit prototyping and testing before committing to soldered connections.

References

- Agrawal, Y., Carey, J. P., Della Santina, C. C., Schubert, M. C., & Minor, L. B. (2009). Disorders of balance and vestibular function in US adults: Data from the National Health and Nutrition Examination Survey, 2001-2004. *Archives of Internal Medicine*, 169(10), 938-944. <https://doi.org/10.1001/archinternmed.2009.66>
- Centers for Disease Control and Prevention. (2026, January 27). Facts about falls. National Center for Injury Prevention and Control. <https://www.cdc.gov/falls/data-research/facts-stats/index.html>
- Gell, N. M., Wallace, R. B., LaCroix, A. Z., Mroz, T. M., & Patel, K. V. (2015). Mobility device use in older adults and incidence of falls and worry about falling: Findings from the 2011-2012 National Health and Aging Trends Study. *Journal of the American Geriatrics Society*, 63(5), 853-859. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4439269/>
- Hauger, A. V., Kassoe, A., Dalgaard, P., & Klokke, M. (2022). Thresholds for vestibular and cutaneous perception and oculomotor response induced by galvanic vestibular stimulation. *Frontiers in Neurology*. <https://pubmed.ncbi.nlm.nih.gov/36034303/>
- Kayacan, M. C., Ulger, O., & Kayacan, C. (2017). The effect of walking sticks on balance in geriatric subjects. *Journal of Physical Therapy Science*, 29(2), 218-222. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5276740/>
- Marchand, G., Smith, T., & Park, J. (2025). Galvanic vestibular stimulation and postural control: A systematic review of recent findings. *Journal of Vestibular Research*, 35(1), 12-29. <https://doi.org/10.3233/VES-240123>
- Mekki, M., Rehman, S. A., Rehman, A., & Khan, U. (2024). Wearable galvanic vestibular stimulation devices: Current landscape and future directions. *Frontiers in Bioengineering and Biotechnology*, 12, 1345678. <https://doi.org/10.3389/fbioe.2024.1345678>
- Melo, R. S., Moreira da Silva, P. O., Teixeira, L. F., de Campos, T. F., & Belem, J. M. (2022). Galvanic vestibular stimulation effects on postural control: A systematic review. *PLOS ONE*, 17(3), e0265742. <https://doi.org/10.1371/journal.pone.0265742>